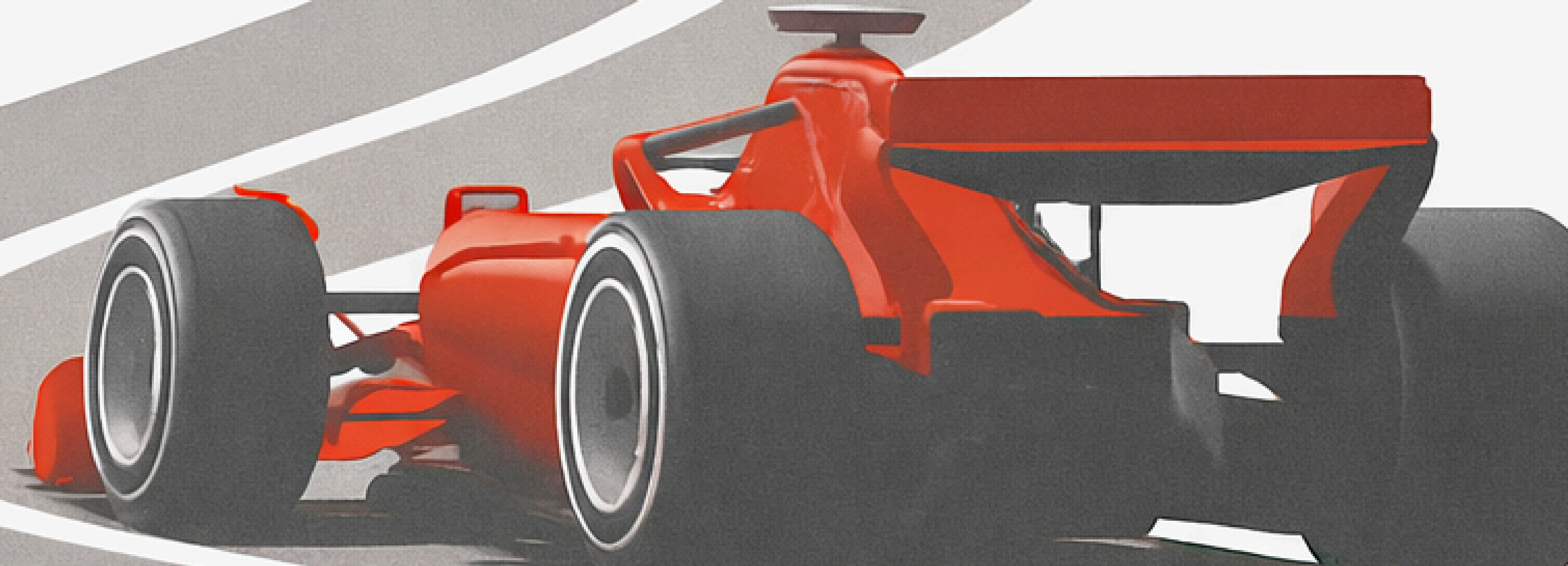

“Not for lap times.
Not for headlines.
For data.”

# F1 Driver Performance Analysis




<u>**Team Principals**</u>

*Carolyn Chang*
*Wendy Cheng*
*Iva Dhooria*
*Kandace Bui*

# Turn 1

## Data & Purpose

# THE DATASET

"F1DriversDataset.csv"

**F1 Drivers Career Statistics**

**1950 to 2023**

**868 Drivers**

**22 Variables**

# THE DATASET

"F1DriversDataset.csv"

## Driver Info

- Name
- Nationality
- Years_Active
- Champion
- Points

## Performance Metrics

- Race_Wins
- Win_Rate
- Podiums
- Podium_Rate
- Pole_Positions
- Fastest_Laps
- Race_Entries
- Points_Per_Entry

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# PROJECT GOAL

Which performance metrics are most predictive of Championship success

Build predictive models to evaluate driver potential

Smarter driver investment decisions

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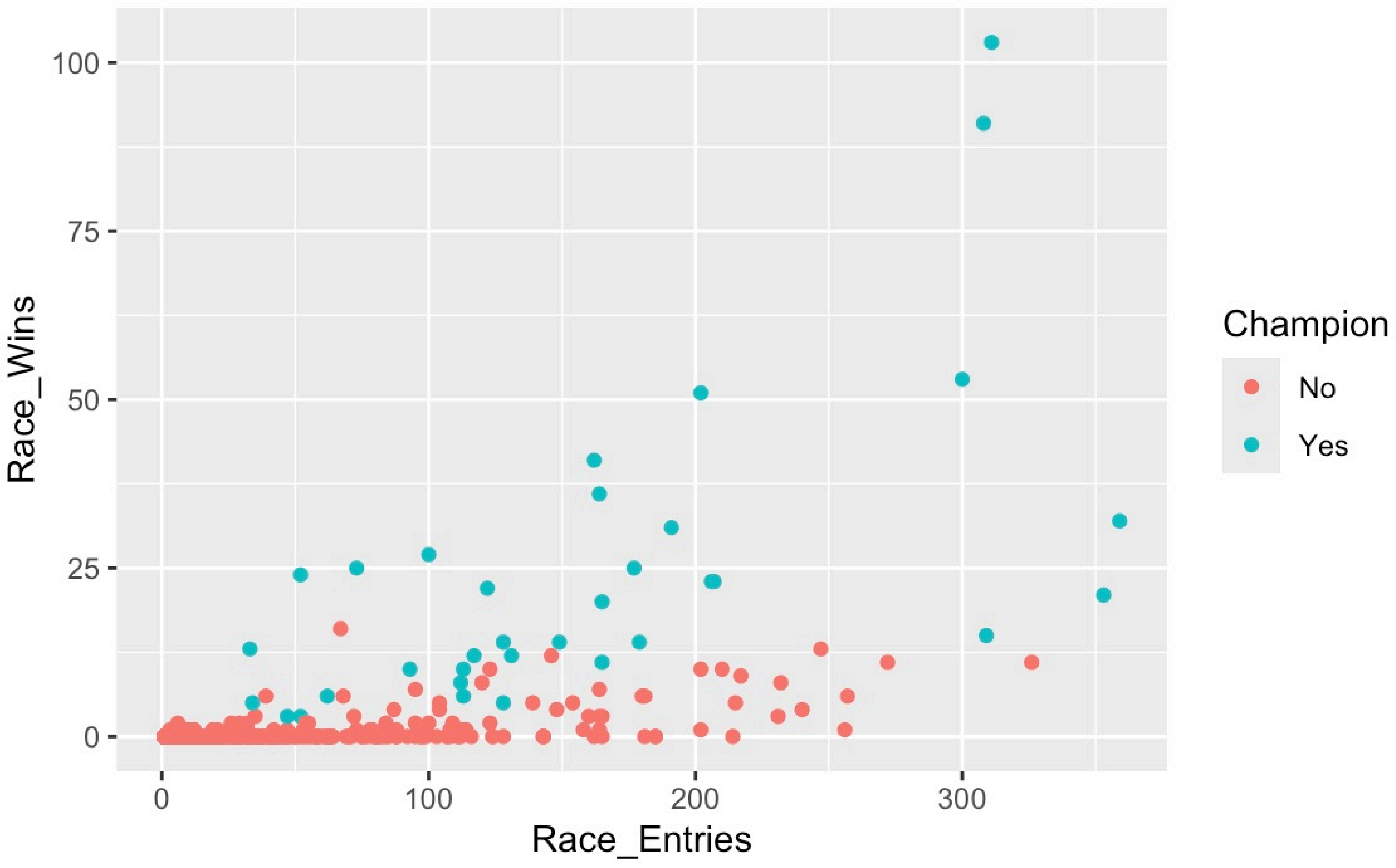
Turn 2
Exploration & Visualization

# SCATTERPLOT

**Scatterplot - Race Wins vs. Race Entries**




**Clear Positive Trend**

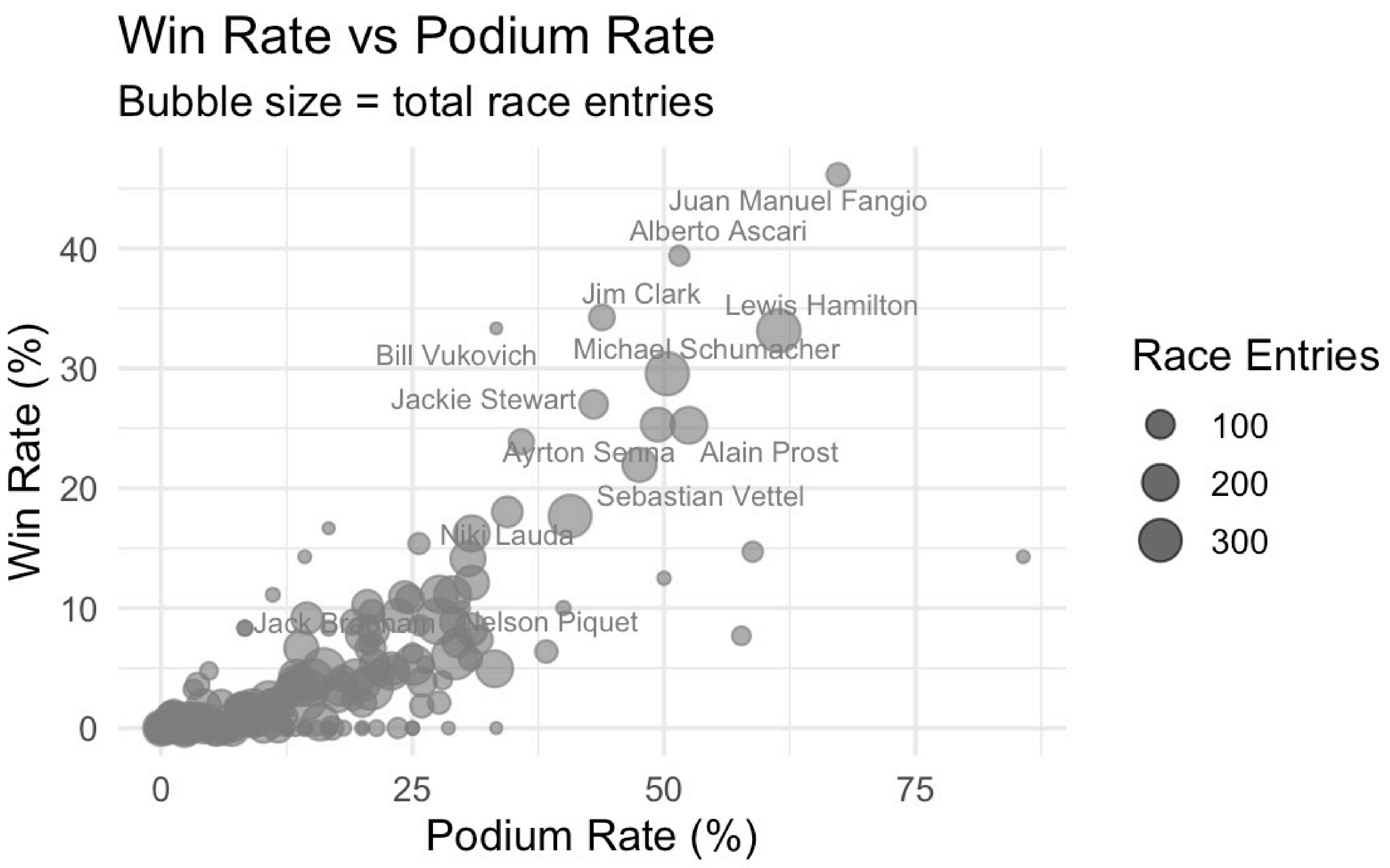
**World champions clustered in upper range of both axes**

**Non-world champions concentrated in zero wins regardless of how many races they enter**

**Race wins is a strong predictor of championship**

# BUBBLE CHART

**Bubble Chart - Win Rate vs. Podium Rate**




**Champion win rate & podium rate:**
**16% & 34%**
**vs.**
**Non-Champion win rate & podium rate:**
**1% & 3%**

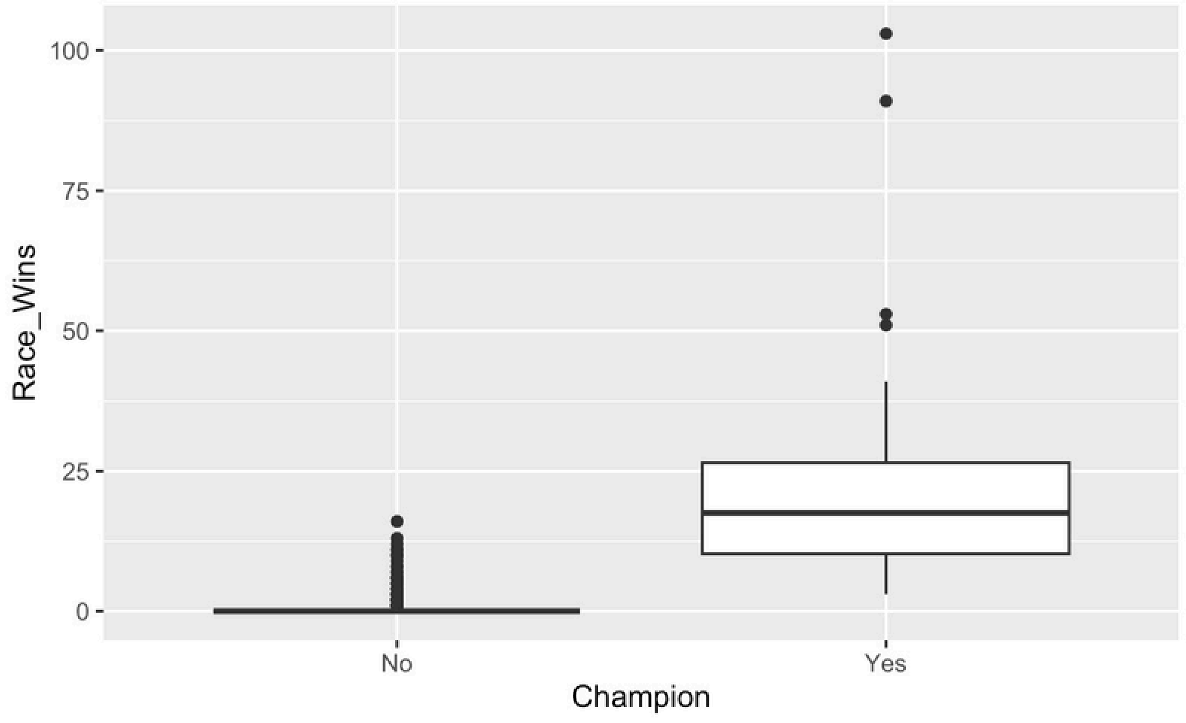
**Win rate & podium rate are both strong predictors**

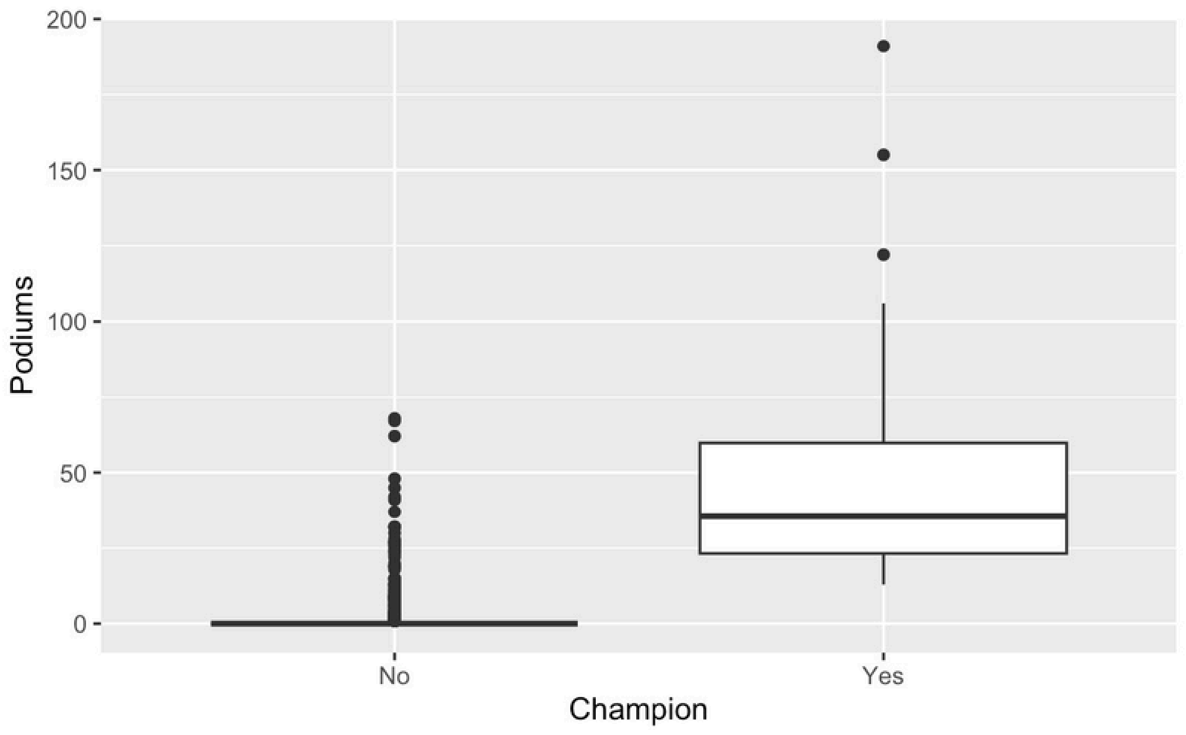
**Difference in race entries & win rate between early era champions vs. modern era champions**

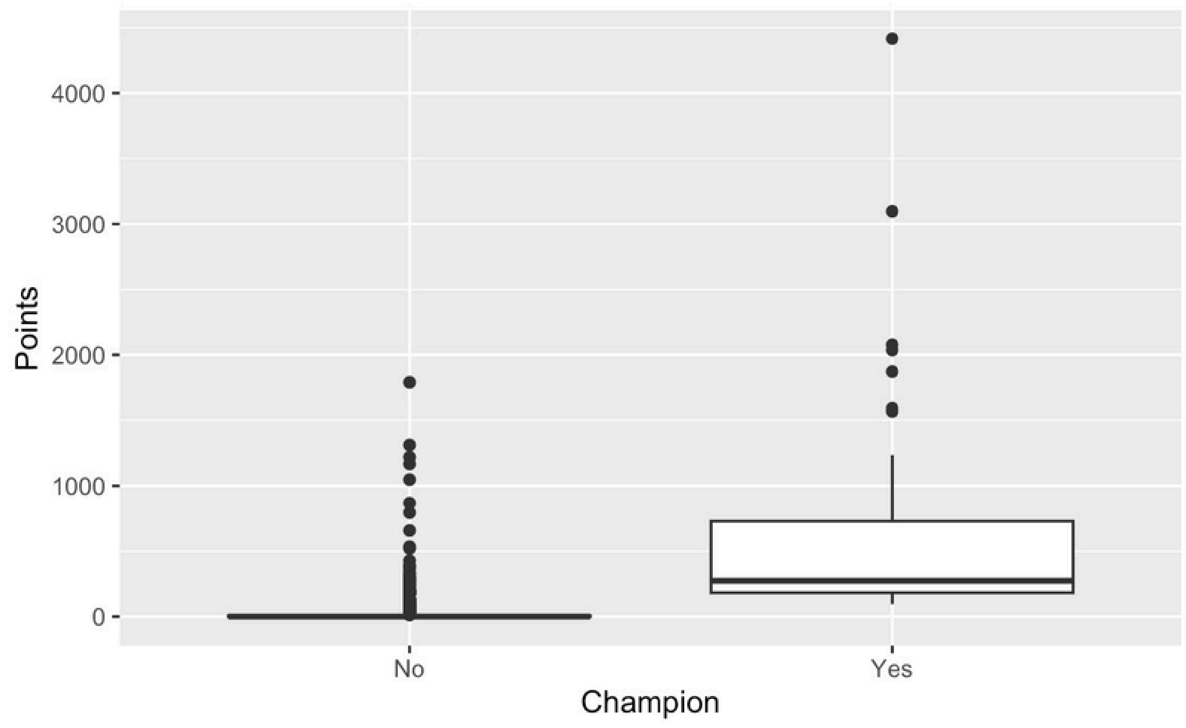
**Rate-based metrics accounting for era effects**

# BOXPLOTS

## Boxplots – Race Wins, Podiums, and Points by Champion Status



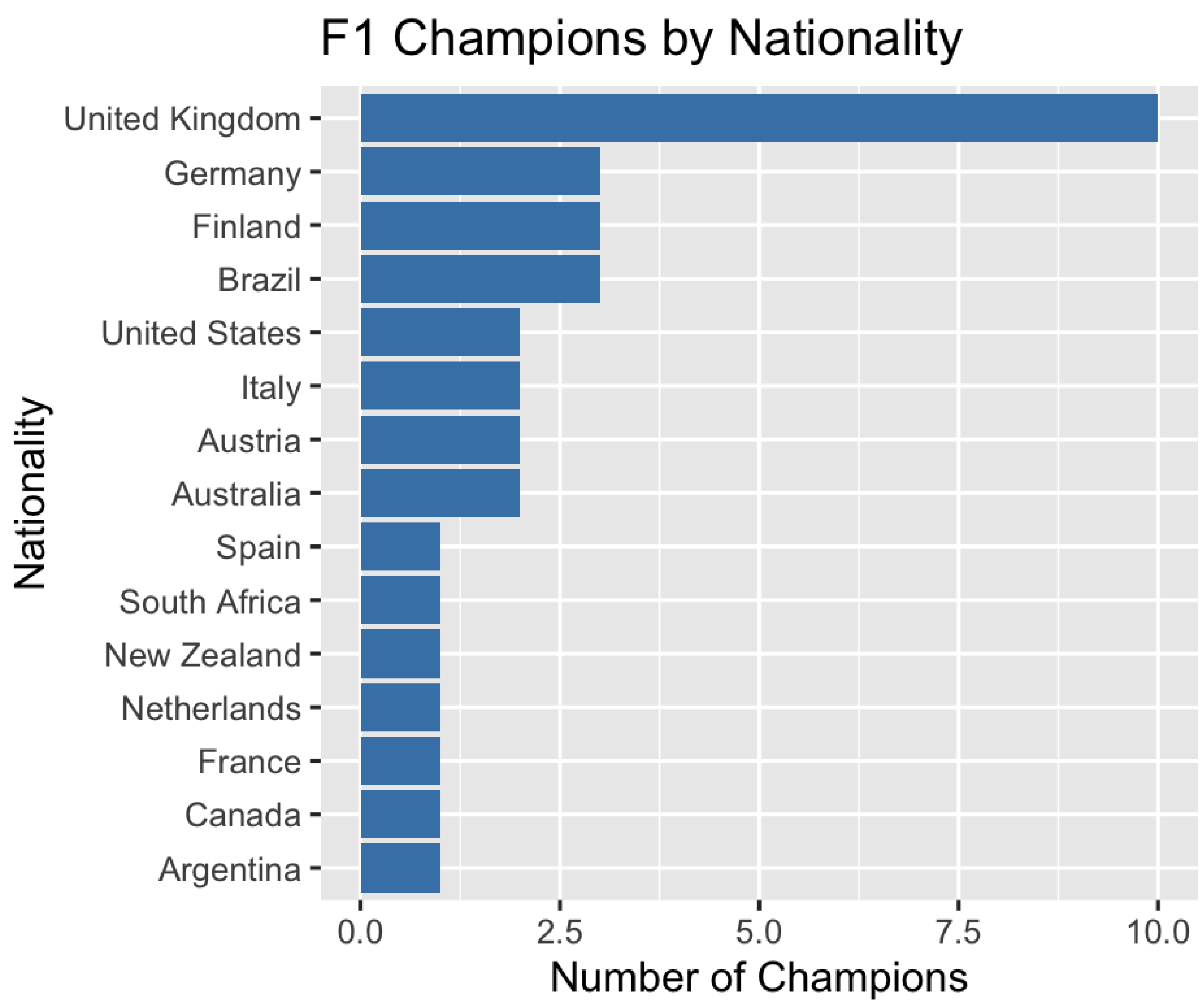

**Champion median**
**Race wins: 17.5**
**Podiums: 35.5**
**Career points: 275.5**
***0 in all three for non-champion***

**Non-champion distributions are heavily right-skewed → most drivers have no wins nor podiums**

**Significant outliers within non-champion: Valtteri Bottas - 1,791 points, 10 wins, no championship**

# BAR CHART

**Bar Chart – F1 Champions by Nationality**

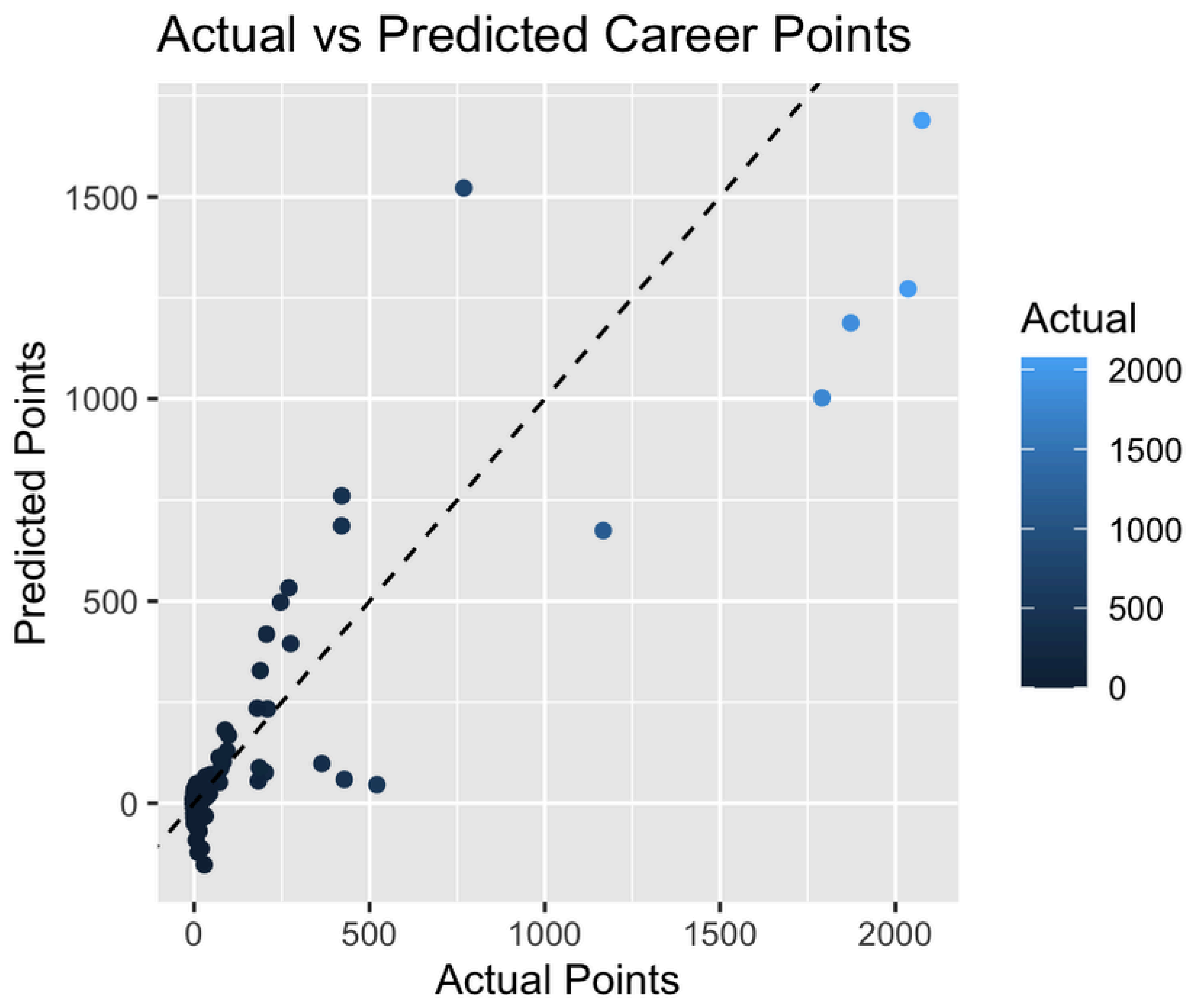



**Provides historical context as of how geographic and resource advantages have shaped driver success**

# Turn 3

## Preprocessing & Mining

# MULTIPLE LINEAR REGRESSION




**R-Squared = 0.812**

**RMSE = 119 and MAE = 42.6**

**Strongest Predictors**
Podiums, Race Entries, Race Wins

**MAE better reflects typical driver predictions, as RMSE is inflated by outliers like Hamilton and Schumacher**

# LOGISTIC REGRESSION

Champion (Yes/No)

Exclude “Points” Variable

70/30 split, 0.5 cutoff

## Key Findings

**Race_Wins**
**Strongest Predictor**
**(p-value = 0.0032)**

**Podium_Rate**
**Marginally Significant**
**(p-value = 0.0577)**

## Model Performance

1. **Accuracy = 98.1%**
2. **Sensitivity = 70%**
3. **Specificity = 99.2%**
4. **Kappa = 0.727**

# CLASSIFICATION

**Champion (Yes/No)**

**Exclude "Points" Variable**

**70/30 split, 0.5 cutoff**

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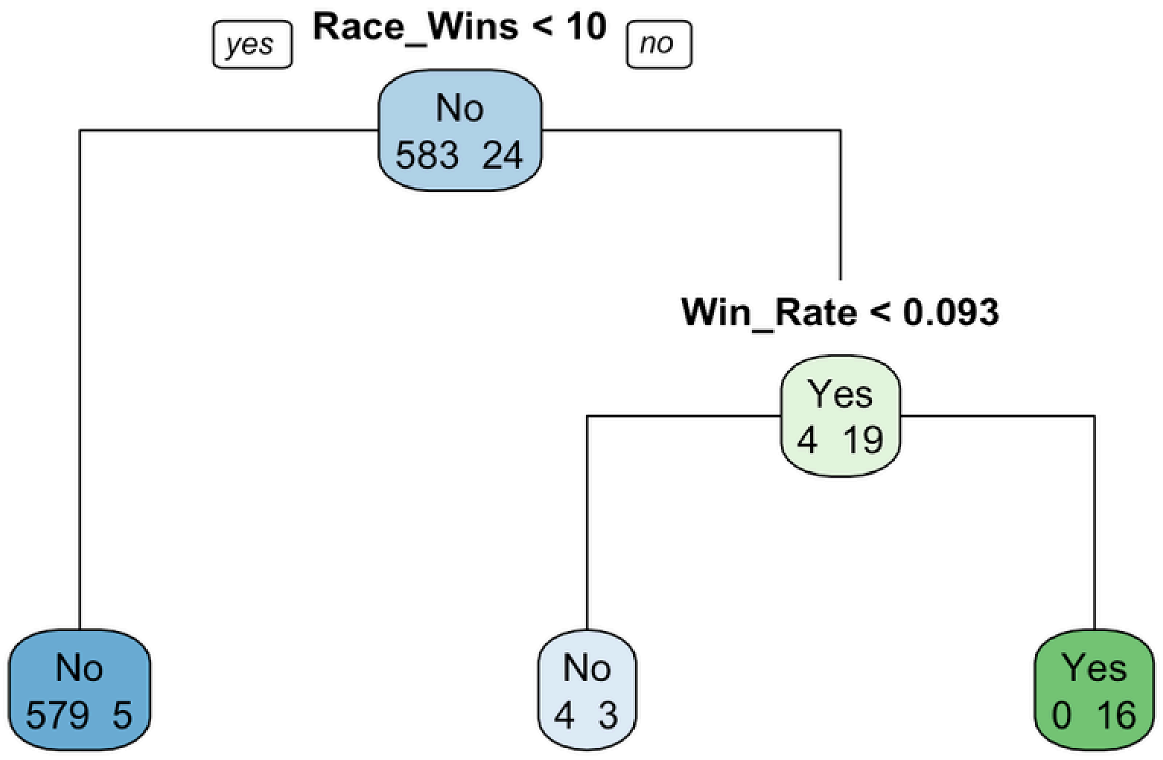
**Classification Tree - F1 Champion Prediction**



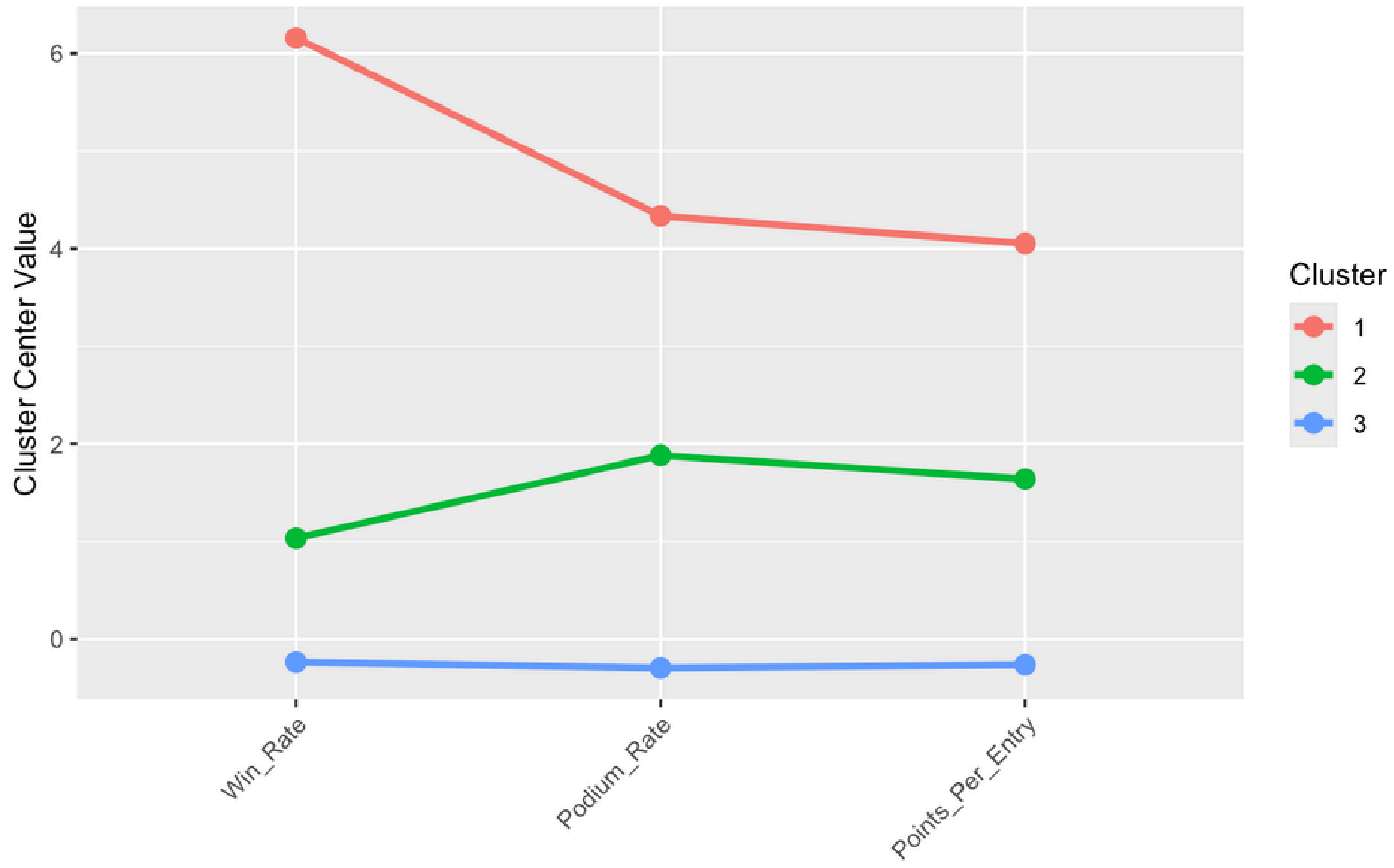

## Model Performance

1. **Accuracy = 97.3%**
2. **Sensitivity = 40%**
3. **Specificity = 99.6%**
4. **Kappa = 0.521**

## Key Takeaway

Logistic Regression outperforms the classification model on sensitivity

# K-MEANS




Win_Rate

Podium_Rate

Points_Per_Entry

Data normalized prior to clustering

Number of clusters (k) = 3

| Tier 1 | Tier 2 | Tier 3 |
| --- | --- | --- |
| *Elite Drivers* | *Competitive Drivers* | *Developing Drivers* |

# Turn 4

## Results & Insights

# RESULTS

## Who Benefits From the Data?

- F1 Team Principals and talent acquisition
- Sponsors and commercial partners
- Junior division racing programs

## How Would the Data Help Them Make Better Decisions?

- Most important metrics: Podiums, Race_Wins
- Career length, pole position, and race volume are not significant predictors

# INSIGHTS

## What Other Data Would Be Useful?

- Car performance metrics
- Other variables: championship standings, car performance rating
- Race-by-race records
- Teammate comparisons
- Driver data: debut age, junior series results, salary

# INSIGHTS

## Obstacles Overcame

1

- Class imbalance in our dataset (Only 34 champions of 868 drivers) → multiple models + sensitivity
- Careful selection of variables to avoid inflating results
- Evaluation of models and model outputs

## Future Improvements

2

- Splitting the dataset by time periods to see F1 evolution
- Incorporating automobile performance metrics
- Removing redundant variables and improving class imbalance

## Role of AI

3

- Chart aesthetics
- Verify and correct R code
- Context for regression model output

# Finish Line

## Thank You! Questions?